

QC/HPC Resource Allocation/Scheduling

Squillante and Tantawi and potentially others

January 2026

1 Introduction

TBA

Our mathematical analysis of parallel computational phases of quantum-centric supercomputing application workloads derives for the first time a closed-form expression for the expected sojourn time of the corresponding fork-join queueing network with nonhomogeneous service rates.

quantum processing units (QPUs) graphical processing units (GPUs) central processing units (CPUs)

The remainder of the paper is organized as follows. Section 2 provides some background and technical preliminaries on quantum-centric supercomputing centers, followed by a presentation of our mathematical models of quantum-centric supercomputing application workloads and systems in Section 3. We first derive in Section 4 closed-form expressions for the expected sojourn time of each of the two basic core components of quantum-centric supercomputing application workloads, followed by a

We then ...

The proofs of our main results are presented in Section ??, followed by concluding remarks.

2 Technical Preliminaries

TBA

3 Mathematical Models

The applications of quantum-centric supercomputing centers are generally characterized by a directed acyclic graph. Here, the directed edges from the set of parent nodes to a child node in the graph define the set of tasks associated with the parent nodes that must be completed before the task associated with the child node can start execution. The basic core components of these directed acyclic graphs that make up quantum-centric supercomputing

application workloads are represented by: (1) a parallel computational phase comprising a single fork-join pair of quantum and classical computational tasks executed in parallel; and (2) a sequential computational phase comprising a single pair of quantum and classical computational tasks executed in series in either order. Then the quantum-centric supercomputing application workloads consist of a mixture of these two basic core components, where quantum computational tasks are executed on system QPUs and classical computational tasks are executed on system GPUs. As a common representative model of these workloads, we shall primarily consider two classes of quantum-centric supercomputing applications with different structures for the computational phases depicted in the directed acyclic graph. The first class, called *parallel-sequential applications*, consists of a parallel computational phase followed by a sequential computational phase. The second class, called *sequential-parallel applications*, consists of a sequential computational phase followed by a parallel computational phase. Although not a strict requirement, we envision that both application classes start with a classical initialization task executed on system CPUs and end with a classical termination task executed on system CPUs.

A job arrives to the quantum-centric supercomputing center according to a Poisson process with rate λ . With probability $p_{\parallel}$, the arriving job belongs to the parallel-sequential application class, and with probability $1 - p_{\parallel}$, the arriving job belongs to the sequential-parallel application class. In both application classes, the sequential computational phase consists of a quantum computational task followed by a classical computational task with probability $p_{\circ}$, and consists of a classical computational task followed by a quantum computational task with probability $1 - p_{\circ}$. The quantum computational tasks of any parallel computational phase or sequential computational phase can involve parallel quantum computations requiring execution on $M_Q \geq 1$ QPUs. Similarly, the classical computational tasks of any parallel computational phase or sequential computational phase can involve parallel classical computations requiring execution on $M_G \geq 1$ GPUs. As our starting point, we shall consider $M_Q = 1$ for all parallel computational phases and $M_G = 1$ for all sequential computational phases. While this helps to reduce the complexity of the mathematical analysis and optimization problems at hand, it is important to note that queueing performance such as the expected sojourn time in a system with M_Q QPUs operating at unit speed converges asymptotically to the corresponding queueing performance in our assumed single QPU with speed M_Q as the system utilization $\rho_Q \rightarrow 1$; and similarly for a system with M_G GPUs operating at unit speed converging asymptotically to our assumed single GPU with speed M_G as the system utilization $\rho_G \rightarrow 1$. (NEED TO FIX THIS. SEE STOYAN.)

The service times for a quantum computational task on system QPUs as part of the parallel-sequential application class follow an exponential distribution with rate μ_Q^P ; similarly, the service times for a quantum computational task on system QPUs as part of the sequential-parallel application class follow an exponential distribution with rate μ_Q^{PS} or μ_Q^{SP} , depending upon the order of the series of tasks in the sequential computational phase. Analogously, the service times for a classical computational task on system GPUs as part of the parallel-sequential application class follow an exponential distribution with rate μ_G^P ; similarly, the service times for a classical computational task on system GPUs as part of the sequential-

parallel application class follow an exponential distribution with rate μ_G^{PS} or μ_G^{SP} , depending upon the order of the series of tasks in the sequential computational phase. The service times for a classical initialization and termination task on system CPUs follow an exponential distribution with rate μ_I and μ_T , respectively, such that $\mu_I, \mu_T \gg \mu_Q^P, \mu_Q^{PS}, \mu_Q^{SP}, \mu_G^P, \mu_G^{PS}, \mu_G^{SP}$. There are three different instances of this quantum-centric supercomputing application workload model. The first, called *balanced application workloads*, consists of homogeneous service rates where $\mu_Q^P = \mu_G^P$, $\mu_Q^{PS} = \mu_G^{PS}$, and $\mu_Q^{SP} = \mu_G^{SP}$; the second instance, called *quantum-dominant application workloads*, consists of nonhomogeneous service rates where $\mu_Q^P < \mu_G^P$, $\mu_Q^{PS} < \mu_G^{PS}$, and $\mu_Q^{SP} < \mu_G^{SP}$; and the third instance, called *classical-dominant application workloads*, consists of nonhomogeneous service rates where $\mu_Q^P > \mu_G^P$, $\mu_Q^{PS} > \mu_G^{PS}$, and $\mu_Q^{SP} > \mu_G^{SP}$. Although all three application workload models are considered in our study, it is important to point out that balanced workloads are very rarely found in quantum-centric supercomputing centers in practice and it is strongly believed that this trend will persist. Quantum-dominant and classical-dominant workloads are expected to be far more prevalent, and thus these two workloads represent an important emphasis in our study.

While our assumptions of Poisson job arrivals and exponential service times help to reduce the complexity of the mathematical analysis and optimization problems at hand, it is critically important to note that there is very little or no data available on the arrival and service patterns for quantum-centric supercomputing centers, and whatever data might be available today is highly unlikely to be representative of future quantum-centric supercomputing application workloads. The field is evolving so quickly with very little in the way of a fundamental analysis of performance in quantum-centric supercomputing centers, making it premature to know the actual distributions of arrival and service patterns in future quantum-centric supercomputing application workloads. Meanwhile, there is an important gap in our understanding of fundamental issues related to design, performance and optimization of quantum-centric supercomputing centers. Given such few studies of this type in the field, we seek to address this gap in the research literature with our study providing the first mathematical analysis and optimization of fundamental performance and related theoretical results in quantum-centric supercomputing centers, to the best of our knowledge.

4 Mathematical Analysis of Core Computational Phases

The two basic core components of quantum-centric supercomputing application workloads consist of either a sequential computational phase or a parallel computational phase. A sequential computational phase comprises with probability p_o a quantum computational task executed on a system QPU followed by a classical computational task executed on a system GPU, and comprises with probability $1 - p_o$ a classical computational task executed on a system GPU followed by a quantum computational task executed on a system QPU. In contrast, a parallel computational phase consists of a single fork-join pair of quantum and classical computational tasks with the former executed on a system QPU and the latter executed on a system GPU. We consider in turn the analysis of each of these two basic core components of quantum-centric supercomputing application workloads in isolation, first

deriving a closed-form expressions for the expected sojourn time of the sequential computational phase and then deriving a closed-form expressions for the expected sojourn time of the parallel computational phase.

4.1 Sequential Computational Phase

The stochastic network corresponding to the sequential computational phase of quantum-centric supercomputing application workloads in isolation represents an open Jackson queueing network [6,17]. Such jobs arrive according to a Poisson process with rate γ , each of which is routed with probability p_o to queue 1 for a system QPU and is routed with probability $1 - p_o$ to queue 2 for a system GPU. Upon completion of the quantum computational task at queue 1, the corresponding jobs are routed directly to queue 3 for a system GPU; and upon completion of the classical computational task at queue 2, the corresponding jobs are routed directly to queue 4 for a system QPU. The arrivals to each queue i follow Poisson processes with rates $\gamma_1 = \gamma_3 = p_o\lambda$ and $\gamma_2 = \gamma_4 = (1 - p_o)\lambda$; and the service times for each queue i follow an exponential distribution with rate $\hat{\mu}_i$, $i \in \{1, \dots, 4\}$, where $\hat{\mu}_1 = S_{Q_1}\mu_{Q_1}$, $\hat{\mu}_2 = S_{G_2}\mu_{G_2}$, $\hat{\mu}_3 = S_{Q_3}\mu_{Q_3}$, $\hat{\mu}_4 = S_{G_4}\mu_{G_4}$. Then, jobs depart the system upon completion of their computational tasks at queue 3 or queue 4. From Burke's Theorem [10] and the quasireversibility of such stochastic networks [18,26], we also have that the departure process from queue 3 and queue 4 are Poisson with rate $p_o\lambda$ and $(1 - p_o)\lambda$, respectively.

Stochastic networks of this form have received considerable attention in the research literature; see, e.g., [3,6,9,17,18,21]. It is generally well known that these open Jackson queueing networks have a product-form solution, which implies that each queue of the stochastic network can be analyzed in a separable manner from the rest of the network. We therefore have that the expected sojourn time at queue i is given by

$$\mathbb{E}[T_i^{PF}] = \frac{1}{\hat{\mu}_i - \gamma_i}, \quad i \in \{1, \dots, 4\}, \quad (1)$$

and that the expected overall sojourn time of the sequential computational phase of quantum-centric supercomputing application workloads in isolation is obtained as the sum of the expected sojourn time at each of the queues i in (1); namely, we have

$$\mathbb{E}[T_{PF}] = \frac{1}{\hat{\mu}_1 - \gamma_1} + \frac{1}{\hat{\mu}_2 - \gamma_2} + \frac{1}{\hat{\mu}_3 - \gamma_3} + \frac{1}{\hat{\mu}_4 - \gamma_4}. \quad (2)$$

Although we primarily focus on a single pair of alternating quantum and classical computational tasks executed in series in either order, the corresponding results for a general number of sequential computational phases would be obtained in a similar manner.

4.2 Parallel Computational Phase

The stochastic network corresponding to the parallel computational phase of quantum-centric supercomputing application workloads in isolation represents a fork-join queueing network

with two parallel servers [13, 14]. Such jobs arrive according to a Poisson process with rate γ , each of which forks into a pair of quantum and classical computational tasks to be executed in parallel with one routed to a queue for the system QPU and the other routed to a queue for the system GPU. The service times for quantum computational tasks at the system QPU queue and for classical computational tasks at the system GPU queue each follow an exponential distribution with rate $\hat{\mu}_Q = S_Q\mu_Q$ and $\hat{\mu}_G = S_G\mu_G$, respectively. Without loss of generality, let $\hat{\mu}_1 = \min\{\hat{\mu}_Q, \hat{\mu}_G\}$ and $\hat{\mu}_2 = \max\{\hat{\mu}_Q, \hat{\mu}_G\}$, and thus $\hat{\mu}_1 \leq \hat{\mu}_2$. Then, jobs depart the system only upon completion of their computational tasks at both queue 1 and queue 2.

Stochastic networks of this form have received considerable attention in the research literature; see, e.g., [5, 13, 14, 25, 27, 33, 37]. In the case of balanced application workloads with homogeneous service rates $\hat{\mu} = \hat{\mu}_1 = \hat{\mu}_2$, it is generally well known that the expected sojourn time of jobs in these fork-join queueing networks is given by

$$\mathbb{E}[T_{FJ}] = \frac{12 - \gamma/\hat{\mu}}{8(\hat{\mu} - \gamma)}, \quad (3)$$

due to Nelson and Tantawi [27]. Hence, the expected overall sojourn time of the parallel computational phase of quantum-centric supercomputing application workloads in isolation can be expressed as in (3).

On the other hand, as discussed in Section 3, the service times of quantum computational tasks and classical computational tasks in quantum-centric supercomputing application workloads are very rarely balanced, if at all, and thus depict either quantum-dominant workloads where $\hat{\mu}_Q < \hat{\mu}_G$ or classical-dominant workloads where $\hat{\mu}_G < \hat{\mu}_Q$. In strong contrast to (3), the corresponding fork-join queueing network under unbalanced workloads with nonhomogeneous service rates generally represents a difficult queueing-theoretic problem for which there is no known exact expression of the expected sojourn time. To address this difficult queueing-theoretic problem, we take a light-heavy traffic interpolation approximation approach which can be an effective method for the analysis of otherwise intractable queueing systems. An overview of this general approach, developed in large part by Reiman and Simon [34–36], is presented in Appendix A. In what follows, we derive our light-heavy traffic interpolation approximation and related theoretical results for two-server fork-join queueing networks with nonhomogeneous service rates, which is of interest in its own right, noting that our approximation yields the exact result in (3) for fork-join queueing networks with homogeneous service rates $\hat{\mu} = \hat{\mu}_1 = \hat{\mu}_2$.

ADD A SUMMARIZED VERSION OF THE DERIVATION AND THEORETICAL RESULTS IN APPENDIX A

5 Mathematical Analysis of Reserved Allocation

We now consider our general quantum-centric supercomputing application and system models of Section 3 under a resource allocation reservation policy in which the resources of the quantum-centric supercomputing center are reserved by jobs in the order of their arrival.

This is a fairly common scheme in practice today for many quantum-centric supercomputing centers. Recall that each arriving job belongs to the parallel-sequential application class with probability $p_{\parallel}$ and belongs to the sequential-parallel application class with probability $1 - p_{\parallel}$, where the former comprises a parallel computational phase followed by a sequential computational phase and the latter comprises a sequential computational phase followed by a parallel computational phase. The sequential computational phase of both application classes consists of a quantum computational task followed by a sequential computational task with probability $p_{\circ}$, and consists of a sequential computational task followed by a quantum computational task with probability $1 - p_{\circ}$. In addition, an arriving job of each application class starts with a classical initialization task and ends with a classical termination task. Due to the fairly common reservation policy of this section, each job reserves the system CPUs, QPUs and GPUs throughout its execution without any overlap with other jobs that wait in the system queue.

More formally, consider a continuous-time Markov process $\{X(t); t \in \mathbb{R}_+\}$ on the state space $\hat{\Omega} = \bigcup_{i=0}^{\infty} \hat{\Omega}_i$ where $\hat{\Omega}_0 := \{0\}$ and $\hat{\Omega}_i := \{(i, j_1, j_2, k, q, g) \mid i \in \mathbb{Z}_+, j_1 \in \{P, S\}, j_2 \in \{P, S, S_{gq}, S_{qg}\}, k \in \{I, C, T\}, q \in \{0, 1\}, g \in \{0, 1\}\}$, for the nonnegative reals $\mathbb{R}_+ := \mathbb{R}_{\geq 0}$ with $\mathbb{R}_{\geq k} := [k, \infty)$, and nonnegative integers $\mathbb{Z}_+ := \mathbb{Z}_{\geq 0}$ with $\mathbb{Z}_{\geq k} := \{k, k+1, k+2, \dots\}$. Here, the state-vector variable i denotes the total number of jobs in the quantum-centric supercomputing center, j_1 denotes whether the job in service belongs to the parallel-sequential application class (P) or the sequential-parallel application class (S), j_2 denotes whether the application in service is executing a parallel computational phase (P) or a sequential computational phase (S) that comprises a classical computational task followed by a quantum computational task (S_{gq}) or a quantum computational task followed by a classical computational task (S_{qg}), k denotes whether the job in service is currently executing its initialization phase (I) or termination phase (T) on the system CPUs or executing its quantum/classical computational phases (C), q denotes whether or not the job in service is currently executing a quantum computational task on the system QPUs, and g denotes whether or not the job in service is currently executing a classical computational task on the system GPUs. We shall refer to the set $\hat{\Omega}_i$ as level i .

Define $[M] := \{1, \dots, M\}$, $M < \infty$. Let $x_{i,m} \in \hat{\Omega}_i$ be a pseudo-lexicographic ordering of the elements of $\hat{\Omega}_i$, $i \in \mathbb{Z}_{\geq 1}$, $m \in [M]$; similarly, let $\hat{\Omega}_0 = \{x_0\}$. The ordering $x_{i,m}$ for level $i \in \mathbb{Z}_{\geq 1}$ is then given by: $(i, P, P, I, 0, 0)$; $(i, P, P, C, 0, 1)$; $(i, P, P, C, 1, 0)$; $(i, P, P, C, 1, 1)$; $(i, P, S_{gq}, C, 0, 1)$; $(i, P, S_{gq}, C, 1, 0)$; $(i, P, S_{gq}, C, 0, 1)$; $(i, P, S_{gq}, C, 1, 0)$; $(i, P, P, T, 0, 0)$; $(i, S, S, I, 0, 0)$; $(i, S, P, C, 0, 1)$; $(i, S, P, C, 1, 0)$; $(i, S, P, C, 1, 1)$; $(i, S, S_{gq}, C, 0, 1)$; $(i, S, S_{gq}, C, 1, 0)$; $(i, S, S_{gq}, C, 0, 1)$; $(i, S, S_{gq}, C, 1, 0)$; $(i, S, S, T, 0, 0)$; and thus $M = 18$. Following this pseudo-lexicographic ordering, the infinitesimal generator matrix $\mathbf{Q}$ of the Markov process $\{X(t); t \in \mathbb{R}_+\}$ has the block tridiagonal structure

$$\mathbf{Q} = \begin{bmatrix} \mathbf{B}_{00} & \mathbf{B}_{01} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots \\ \mathbf{B}_{10} & \mathbf{B}_{11} & \mathbf{A}_1 & \mathbf{0} & \mathbf{0} & \cdots \\ \mathbf{0} & \mathbf{A}_{-1} & \mathbf{A}_0 & \mathbf{A}_1 & \mathbf{0} & \cdots \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_{-1} & \mathbf{A}_0 & \mathbf{A}_1 & \ddots \\ \vdots & \vdots & \ddots & \ddots & \ddots & \ddots \end{bmatrix}, \quad (4)$$

where $\mathbf{B}_{00}$, $\mathbf{B}_{01}$, $\mathbf{B}_{10}$, $\mathbf{B}_{11}$ and $\mathbf{A}_i$, $i \in \{-1, 0, 1\}$, are finite matrices of dimensions 1×1 , $1 \times M$, $M \times 1$, $M \times M$ and $M \times M$, respectively. By definition, since $\mathbf{Q}$ is a generator matrix, we have that $\mathbf{B}_{01}$, $\mathbf{B}_{10}$, $\mathbf{A}_{-1}$ and $\mathbf{A}_1$ are nonnegative matrices while $\mathbf{B}_{00}$, $\mathbf{B}_{11}$ and $\mathbf{A}_0$ have nonnegative offdiagonal elements and strictly negative diagonal elements such that the row sums of $\mathbf{Q}$ equal 0. Moreover, we have that $\mathbf{B}_0 = \mathbf{B}_{00} + \mathbf{B}_{01}$, $\mathbf{B}_1 = \mathbf{B}_{10} + \mathbf{B}_{11} + \mathbf{A}_1$, and $\mathbf{A} = \mathbf{A}_{-1} + \mathbf{A}_0 + \mathbf{A}_1$ are generator matrices.

The matrices corresponding to the non-homogeneous boundary of the state space $\hat{\Omega}$ are given by

$$\mathbf{B}_{00} = [-\lambda], \quad \mathbf{B}_{01} = [\mathbf{B}_{01}^P \quad \mathbf{B}_{01}^S], \quad \mathbf{B}_{10} = [\mathbf{B}_{10}^P \quad \mathbf{B}_{10}^S]^\top, \quad (5)$$

$$\mathbf{B}_{01}^P = [p_{||}\lambda \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0], \quad \mathbf{B}_{01}^S = [(1-p_{||})\lambda \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0], \quad (6)$$

$$\mathbf{B}_{10}^P = [0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad \mu_T^P], \quad \mathbf{B}_{10}^S = [0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad \mu_T^S], \quad (7)$$

$$\mathbf{B}_{11} = \begin{bmatrix} \mathbf{B}_{11}^P & \mathbf{0} \\ \mathbf{0} & \mathbf{B}_{11}^S \end{bmatrix}, \quad \mathbf{B}_{11}^P = \begin{bmatrix} \mathbf{B}_{11}^{P_{pp}} & \mathbf{B}_{11}^{P_{ps}} \\ \mathbf{0} & \mathbf{B}_{11}^{P_{ss}} \end{bmatrix}, \quad \mathbf{B}_{11}^S = \begin{bmatrix} \mathbf{B}_{11}^{S_{pp}} & \mathbf{B}_{11}^{S_{ps}} \\ \mathbf{B}_{11}^{S_{sp}} & \mathbf{B}_{11}^{S_{ss}} \end{bmatrix}, \quad (8)$$

$$\mathbf{B}_{11}^{P_{pp}} = \begin{bmatrix} (\lambda, \mu_I^P) & 0 & 0 & \mu_I^P \\ 0 & (\lambda, \mu_G^{PP}) & 0 & 0 \\ 0 & 0 & (\lambda, \mu_Q^{PP}) & 0 \\ 0 & \mu_Q^{PP} & \mu_G^{PP} & (\lambda, \mu_Q^{PP}, \mu_G^{PP}) \end{bmatrix}, \quad (9)$$

$$\mathbf{B}_{11}^{P_{ps}} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ (1-p_{\circ})\mu_G^{PP} & 0 & p_{\circ}\mu_G^{PP} & 0 \\ (1-p_{\circ})\mu_Q^{PP} & 0 & p_{\circ}\mu_Q^{PP} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad (10)$$

$$\mathbf{B}_{11}^{P_{ss}} = \begin{bmatrix} (\lambda, \mu_G^{PS}) & \mu_G^{PS} & 0 & 0 & 0 \\ 0 & (\lambda, \mu_Q^{PS}) & 0 & 0 & \mu_Q^{PS} \\ 0 & 0 & (\lambda, \mu_G^{PS}) & 0 & \mu_G^{PS} \\ 0 & 0 & \mu_Q^{PS} & (\lambda, \mu_Q^{PS}) & 0 \\ 0 & 0 & 0 & 0 & (\lambda, \mu_T^P) \end{bmatrix}, \quad (11)$$

$$\mathbf{B}_{11}^{S_{pp}} = \begin{bmatrix} (\lambda, \mu_I^S) & 0 & 0 & 0 \\ 0 & (\lambda, \mu_G^{SP}) & 0 & 0 \\ 0 & 0 & (\lambda, \mu_Q^{SP}) & 0 \\ 0 & \mu_Q^{SP} & \mu_G^{SP} & (\lambda, \mu_G^{SP}, \mu_Q^{SP}) \end{bmatrix}, \quad (12)$$

$$\mathbf{B}_{11}^{S_{ps}} = \begin{bmatrix} (1-p_{\circ})\mu_I^S & 0 & 0 & p_{\circ}\mu_I^S & 0 \\ 0 & 0 & 0 & 0 & \mu_G^{SP} \\ 0 & 0 & 0 & 0 & \mu_Q^{SP} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (13)$$

$$\mathbf{B}_{11}^{S_{sp}} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \mu_Q^{SS} \\ 0 & 0 & 0 & \mu_G^{SS} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{B}_{11}^{S_{ss}} = \begin{bmatrix} (\lambda, \mu_G^{SS}) & \mu_G^{SS} & 0 & 0 & 0 \\ 0 & (\lambda, \mu_Q^{SS}) & 0 & 0 & 0 \\ 0 & 0 & (\lambda, \mu_G^{SS}) & 0 & 0 \\ 0 & 0 & \mu_Q^{SS} & (\lambda, \mu_Q^{SS}) & 0 \\ 0 & 0 & 0 & 0 & (\lambda, \mu_T^S) \end{bmatrix}, \quad (14)$$

where $(x, y) = -(x + y)$ and $(x, y, z) = -(x + y + z)$. Here, the matrix $\mathbf{B}_{10}$ defines the rate of transitions from states in $\hat{\Omega}_1$ to the state $\hat{\Omega}_0$, the matrix $\mathbf{B}_{01}$ defines the rate of transitions from the state $\hat{\Omega}_0$ to states in $\hat{\Omega}_1$, and the offdiagonal elements of the matrix $\mathbf{B}_{11}$ define the rate of transitions among states in $\hat{\Omega}_1$. Similarly, for the homogeneous portion of the state space $\hat{\Omega}$, the matrix $\mathbf{A}_{-1}$ defines the rate of transitions from states in $\hat{\Omega}_{i+1}$ to states in $\hat{\Omega}_i$, the matrix $\mathbf{A}_1$ defines the rate of transitions from states in $\hat{\Omega}_i$ to states in $\hat{\Omega}_{i+1}$, and the offdiagonal elements of the matrix $\mathbf{A}_0$ define the rate of transitions among states in $\hat{\Omega}_i$, $i \in \mathbb{Z}_{\geq 1}$. Hence, the matrices for the homogeneous portion of the state space are given by $\mathbf{A}_1 = \lambda \mathbf{I}$, $\mathbf{A}_0 = \mathbf{B}_{11}$, and

$$\mathbf{A}_{-1} = \begin{bmatrix} \mathbf{A}_{-1}^{PP} & \mathbf{A}_{-1}^{PS} \\ \mathbf{A}_{-1}^{SP} & \mathbf{A}_{-1}^{SS} \end{bmatrix}, \quad (15)$$

$$\mathbf{A}_{-1}^{PP} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ p_{\parallel} \mu_T^P & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{A}_{-1}^{PS} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ (1 - p_{\parallel}) \mu_T^P & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (16)$$

$$\mathbf{A}_{-1}^{SP} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ p_{\parallel} \mu_T^S & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{A}_{-1}^{SS} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ (1 - p_{\parallel}) \mu_T^S & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (17)$$

where $\mathbf{I}$ denotes the identity matrix of appropriate dimension.

Following the pseudo-lexicographic ordering of the elements of each level i of the state space, define

$$\boldsymbol{\pi} := (\boldsymbol{\pi}_0, \boldsymbol{\pi}_1, \boldsymbol{\pi}_2, \dots), \quad (18)$$

$$\boldsymbol{\pi}_0 := (\hat{\pi}(x_0)), \quad \hat{\pi}(x_0) := \lim_{t \rightarrow \infty} \mathbb{P}[X(t) = x_0], \quad (19)$$

$$\boldsymbol{\pi}_i := (\hat{\pi}(x_{i,1}), \dots, \hat{\pi}(x_{i,M})), \quad \hat{\pi}(x_{i,m}) := \lim_{t \rightarrow \infty} \mathbb{P}[X(t) = x_{i,m}], \quad i \in \mathbb{Z}_{\geq 1}, \quad m \in [M]. \quad (20)$$

Assuming the Markov process $\{X(t); t \in \mathbb{R}_+\}$ to be irreducible and ergodic, the limiting probability vector $\boldsymbol{\pi}$ is the stationary distribution of this stochastic process such that its

invariant probability vector $\boldsymbol{\pi}$ exists and is uniquely determined as the solution of $\boldsymbol{\pi} = \boldsymbol{\pi}\mathbf{P}$ and $\boldsymbol{\pi}\mathbf{e} = 1$, where $\mathbf{e} = (1, \dots, 1)^\top$ [9, 22]. Let $\text{sp}(\mathbf{Y})$ denote the spectral radius of a matrix $\mathbf{Y}$, and let a solution $\mathbf{Y}$ be called a minimal nonnegative solution of a matrix equation if $\mathbf{0} \leq \mathbf{Y} \leq \mathbf{Z}$ for any other nonnegative solution $\mathbf{Z}$, with the inequalities holding componentwise. The stationary probability vector $\boldsymbol{\pi}$ of the process $\{X(t); t \in \mathbb{R}_+\}$ with generator $\mathbf{Q}$ then can be obtained from the following matrix-analytic results [Theorems 1.7.1 and 1.5.1] [29].

Theorem 1. *Suppose that the Markov process $\{X(t); t \in \mathbb{R}_+\}$ is irreducible and aperiodic with $\mathbf{Q}$ in the form of (4). Further suppose the generator matrix $\mathbf{A} = \mathbf{A}_{-1} + \mathbf{A}_0 + \mathbf{A}_1$ is irreducible. Then this stochastic process is positive recurrent if and only if $\text{sp}(\mathbf{R}) < 1$, where $\mathbf{R}$ is the minimal nonnegative solution of the quadratic matrix equation*

$$\mathbf{R}^2\mathbf{A}_{-1} + \mathbf{R}\mathbf{A}_0 + \mathbf{A}_1 = \mathbf{0}. \quad (21)$$

Furthermore, there exists a positive vector $(\boldsymbol{\pi}_0, \boldsymbol{\pi}_1)$ such that

$$(\boldsymbol{\pi}_0, \boldsymbol{\pi}_1) \begin{bmatrix} \mathbf{B}_{00} & \mathbf{B}_{01} \\ \mathbf{B}_{10} & \mathbf{B}_{11} + \mathbf{R}\mathbf{A}_{-1} \end{bmatrix} = \mathbf{0}. \quad (22)$$

The remaining elements of the invariant probability vector $\boldsymbol{\pi}$ are then given by

$$\boldsymbol{\pi}_{n+1} = \boldsymbol{\pi}_1\mathbf{R}^n, \quad n \in \mathbb{Z}_+, \quad (23)$$

where $(\boldsymbol{\pi}_0, \boldsymbol{\pi}_1)$ is normalized by

$$\boldsymbol{\pi}_0 + \boldsymbol{\pi}_1(\mathbf{I} - \mathbf{R})^{-1}\mathbf{e} = 1. \quad (24)$$

We have above assumed the irreducibility and ergodicity of the Markov process $\{X(t); t \in \mathbb{R}_+\}$. It readily follows from the definition of $\{X(t); t \in \mathbb{R}_+\}$ and its generator matrix $\mathbf{Q}$ that the Markov process is irreducible and aperiodic. Then the stability conditions for the Markov process to be positive recurrent are provided in Theorem 1, under the additional assumption that the generator matrix $\mathbf{A} = \mathbf{A}_{-1} + \mathbf{A}_0 + \mathbf{A}_1$ is irreducible. The irreducibility of $\mathbf{A}$ can be readily verified, as the directed graph corresponding to $\mathbf{A}$ is strongly connected with transitions from state $(*, P, P, I, 0, 0)$ to state $(*, P, P, T, 0, 0)$ reaching every state $(*, P, *, C, *, *)$ in between, transitions from state $(*, S, S, I, 0, 0)$ to state $(*, S, S, T, 0, 0)$ reaching every state $(*, S, *, C, *, *)$ in between, and transitions from states $(*, *, *, T, 0, 0)$ to states $(*, *, *, I, 0, 0)$ connecting the states across both application classes. Since the generator matrix $\mathbf{A}$ of the Markov process $\{X(t); t \in \mathbb{R}_+\}$ is irreducible, we then have necessary and sufficient conditions for $\text{sp}(\mathbf{R}) < 1$ from the following matrix-analytic result (see Theorem (TO BE LOOKED UP) in [29]). The stochastic process is therefore positive recurrent if and only if (25) is satisfied.

Theorem 2. *Suppose that the Markov process $\{X(t); t \in \mathbb{R}_+\}$ is irreducible. Further suppose that the generator $\mathbf{A}$ is irreducible. Then $\text{sp}(\mathbf{R}) < 1$ if and only if*

$$\mathbf{y}\mathbf{A}_1\mathbf{e} < \mathbf{y}\mathbf{A}_{-1}\mathbf{e}, \quad (25)$$

where $\mathbf{y}$ is the stationary probability vector of $\mathbf{A}$, i.e., the (unique) solution of $\mathbf{y}\mathbf{A} = \mathbf{0}$ and $\mathbf{y}\mathbf{e} = 1$.

Once the stationary distribution is determined, various functions of the elements of the invariant probability vector $\boldsymbol{\pi}$ can be used to obtain performance measures and quantities of interest. In particular, the tail distribution of the queue length process can be expressed as

$$\mathbb{P}[Q > z] = \sum_{n=z+1}^{\infty} \pi_n \mathbf{e} = \boldsymbol{\pi}_1 \mathbf{R}^z (\mathbf{I} - \mathbf{R})^{-1} \mathbf{e}, \quad z \in \mathbb{Z}_+, \quad (26)$$

with the corresponding expectation given by

$$\mathbb{E}[Q] = \sum_{n=0}^{\infty} (n+1) \pi_{n+1} \mathbf{e} = \boldsymbol{\pi}_1 (\mathbf{I} - \mathbf{R})^{-1} \mathbf{e} + \boldsymbol{\pi}_1 \mathbf{R} (\mathbf{I} - \mathbf{R})^{-2} \mathbf{e}. \quad (27)$$

The expected sojourn time of a job can then be calculated using Little's law [45] and (27), which yields

$$\mathbb{E}[T] = \lambda^{-1} \left(\boldsymbol{\pi}_1 (\mathbf{I} - \mathbf{R})^{-1} \mathbf{e} + \boldsymbol{\pi}_1 \mathbf{R} (\mathbf{I} - \mathbf{R})^{-2} \mathbf{e} \right). \quad (28)$$

The asymptotic decay rate of the tail distribution of the queue length process is another measure of interest. Let η denote this decay rate, which is also called the caudal characteristic [31, 32]. Due to the irreducibility of the Markov process $\{X(t); t \in \mathbb{R}_+\}$ and the irreducibility of the generator matrix $\mathbf{A}$, we either have $\eta = \text{sp}(\mathbf{R})$ when the matrix $\mathbf{R}$ is irreducible, or otherwise have $\eta = \text{sp}(\mathbf{R}_{11})$ where, after appropriate row and column permutations,

$$\mathbf{R} = \begin{bmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{0} & \mathbf{R}_{22} \end{bmatrix}$$

with $\mathbf{R}_{11}$ irreducible and $\mathbf{R}_{22}$ strictly upper triangular. We then conclude that $\mathbf{R}^z = O(\eta^z)$, as $z \rightarrow \infty$. Let $\mathbf{u}$ and $\mathbf{v}$ be the left and right eigenvectors corresponding to η that are strictly positive and normalized by $\mathbf{u}\mathbf{e} = 1$ and $\mathbf{u}\mathbf{v} = 1$. From the properties of the Markov process $\{X(t); t \in \mathbb{R}_+\}$, we know that [38]

$$\mathbf{R}^z = \eta^z \mathbf{v} \cdot \mathbf{u} + o(\eta^z), \quad \text{as } z \rightarrow \infty,$$

which together with equation (23) yields

$$\boldsymbol{\pi}_{z+1} \mathbf{e} = \boldsymbol{\pi}_1 \mathbf{v} \eta^z + o(\eta^z), \quad \text{as } z \rightarrow \infty. \quad (29)$$

It then follows that

$$\mathbb{P}[Q \geq z+1] = \frac{\boldsymbol{\pi}_1 \mathbf{v}}{1 - \eta} \eta^z + o(\eta^z), \quad \text{as } z \rightarrow \infty, \quad (30)$$

and thus

$$\lim_{z \rightarrow \infty} \frac{\mathbb{P}[Q \geq z+1]}{\eta^z} = \frac{\boldsymbol{\pi}_1 \mathbf{v}}{1 - \eta}, \quad (31)$$

or equivalently

$$\mathbb{P}[Q \geq z + 1] \sim \frac{\pi_1 \mathbf{v}}{1 - \eta} \eta^z, \quad \text{as } z \rightarrow \infty, \quad (32)$$

where $f(z) \sim g(z)$ denotes that $\lim_{z \rightarrow \infty} f(z)/g(z) = 1$. Hence, in addition to providing the stability criterion for the quantum-centric supercomputing center, η is indicative of the tail behavior of the stationary queue length distribution.

Note that the caudal characteristic can be obtained without having to first solve for the matrix $\mathbf{R}$ in (21). Define the matrix $\tilde{\mathbf{A}}(s) := \mathbf{A}_1 + s\mathbf{A}_0 + s^2\mathbf{A}_{-1}$, for $0 < s \leq 1$. When the generator matrix $\mathbf{A}$ is irreducible, this matrix $\tilde{\mathbf{A}}(s)$ is irreducible with nonnegative off-diagonal elements. Let $\tilde{\eta}(s)$ denote the spectral radius of the matrix $\tilde{\mathbf{A}}(s)$. Then, under the above assumptions, η is the unique solution in $(0,1)$ of the equation $\tilde{\eta}(s) = 0$. This solution can be directly computed, although more efficient methods have been developed [7].

Another set of performance metrics of interest concern the stationary utilization of the system CPUs, QPUs and GPUs composing the quantum-centric supercomputing center. For each level i , $i \in \mathbb{Z}_+$, the states with elements $k \in \{I, T\}$ represent execution on the system CPUs, the states with elements $q = 1$ represent execution on the system QPUs, and the states with elements $g = 1$ represent execution on the system GPUs. The corresponding stationary utilizations can then be expressed as

$$U_C = \sum_{n=0}^{\infty} \pi_{n+1} \nu_C = \pi_1 (\mathbf{I} - \mathbf{R})^{-1} \nu_C, \quad (33)$$

$$U_Q = \sum_{n=0}^{\infty} \pi_{n+1} \nu_Q = \pi_1 (\mathbf{I} - \mathbf{R})^{-1} \nu_Q, \quad (34)$$

$$U_G = \sum_{n=0}^{\infty} \pi_{n+1} \nu_G = \pi_1 (\mathbf{I} - \mathbf{R})^{-1} \nu_G, \quad (35)$$

respectively where the ℓ^{th} element of the vector ν_C contains a 1 if $k \in \{I, T\}$ in the corresponding state of the homogeneous portion and contains a 0 otherwise, the ℓ^{th} element of the vector ν_Q contains a 1 if $q = 1$ in the corresponding state of the homogeneous portion and contains a 0 otherwise, and the ℓ^{th} element of the vector ν_G contains a 1 if $g = 1$ in the corresponding state of the homogeneous portion and contains a 0 otherwise.

6 Optimization of Resource Allocation

In contrast to the previous section, as an alternative to the resource allocation reservation policy in fairly common practice today for many quantum-centric supercomputing centers, we shall consider in this section a general stochastic network that explicitly represents the queues associated with the quantum-centric supercomputing application and system models of Section 3. A primary goal of this paper is to determine the optimal allocation of resource capacity within the context of quantum-centric supercomputing centers. The fundamental tradeoff here is between the performance received by jobs submitted to the high-performance

computing center and the financial concerns incurred by the host of the center. We shall focus on the expected sojourn times at the queues of the stochastic network as the performance metric of interest and shall focus on the costs of the resource capacities as the financial metric of interest, noting that related performance and financial metrics can be addressed in a similar manner. Since some queues of the stochastic network may be more important than others, we shall also allow a weight to be assigned to each queue. Then, our formulation of the resource capacity allocation optimization problem of interest consists of minimizing the sum of the weighted expected sojourn times in the stochastic network subject to a budgetary cost constraint on the resource capacities and to the constraint that the stochastic network is stable. Alternatively, we could similarly consider an optimization formulation based on minimizing the cost of the resource capacities subject to a constraint on the sum of the weighted expected sojourn times.

More formally, let N denote the number of queues comprising the stochastic network with any ordering of interest for these queues. Define the vectors $\boldsymbol{\gamma} := (\gamma_1, \dots, \gamma_N)$ and $\hat{\boldsymbol{\mu}} := (\hat{\mu}_1, \dots, \hat{\mu}_N)$, where γ_i is the effective arrival rate to queue i and $\hat{\mu}_i$ is the resource capacity of queue i . Let $\mathcal{C}$ denote the total budgetary cost constraint on the resource capacities, and $\mathbb{E}[T_i]$ the expected sojourn time at queue i , $i \in [N]$. Further define the vectors $\mathbf{c} := (c_1, \dots, c_N)$ and $\boldsymbol{\omega} := (\omega_1, \dots, \omega_N)$, where c_i is the cost associated with each unit of resource capacity $\hat{\mu}_i$ at queue i and ω_i is the weight associated with the expected sojourn time at queue i , $i \in [N]$. Our resource capacity allocation optimization problem is then formulated as follows.

$$\text{(RCA - OPT)} \quad \min_{\hat{\boldsymbol{\mu}} \in (0, \infty)^N} \sum_{i=1}^N \omega_i \mathbb{E}[T_i] \quad (36)$$

$$\text{s.t.} \quad \langle \mathbf{c}, \hat{\boldsymbol{\mu}} \rangle \leq \mathcal{C}, \quad \text{and} \quad \hat{\mu}_i > \gamma_i, \quad \forall i \in [N]. \quad (37)$$

The difficulty in solving (RCA-OPT) is rooted in determining the expected sojourn times $\mathbb{E}[T_i]$ as a function of the resource capacity vector $\hat{\boldsymbol{\mu}}$. As described in Section 4.1, we know that the sequential computational phase represents a product-form queueing network. In addition, a sequential computational phase followed by the fork-join queueing network of a parallel computational phase would also represent a product-form queueing network, given the Poisson departure process from the sequential computational phase due to Burke's Theorem [10] and quasireversibility [18, 26]; see Section 4.1. However, even though the service times at each queue follow an exponential distribution, the general application workload model comprising combinations of both the parallel-sequential application class and the sequential-parallel application class does not ensure Poisson arrival processes to each queue of the stochastic network. The main difficulty is not knowing the higher-order properties of the arrival process to each queue i beyond its effective arrival γ_i ; moreover, these properties of the arrival process depend on the resource capacities of the queues j that precede queue i in the stochastic network, $i, j \in [N]$. Hence, a standard approach for solving optimization problems such as (RCA-OPT) is based on a combination of simulation methods (to estimate the $\mathbb{E}[T_i]$ for a given $\hat{\boldsymbol{\mu}}$) and gradient methods (to obtain the optimal service rates), such as stochastic approximation [1, 4, 19].

To address the computational complexity of stochastic approximation, we follow the approach proposed by Dieker, Ghosh and Squillante [11] within the context of quantum-centric supercomputing centers. Observe from (1) and (3), and even (61), that the expression for the expected sojourn time of queue i under any resource capacity allocation $\hat{\boldsymbol{\mu}}$ has a functional form $\mathbb{E}[T_i] = \zeta_i/(\hat{\mu}_i - \gamma_i)$, $i \in [N]$. Upon substituting this functional form into the objective function (36) of (RCA-OPT), we obtain the revised optimization problem formulation

$$\text{(RCA - OPT : FF)} \quad \min_{\hat{\boldsymbol{\mu}} \in (0, \infty)^N} \sum_{i=1}^N \omega_i \frac{\zeta_i}{\hat{\mu}_i - \gamma_i} \quad (38)$$

$$\text{s.t.} \quad \langle \mathbf{c}, \hat{\boldsymbol{\mu}} \rangle \leq \mathcal{C}, \quad \text{and} \quad \hat{\mu}_i > \gamma_i, \quad \forall i \in [N]. \quad (39)$$

Then, given the vector $\boldsymbol{\zeta} := (\zeta_1, \dots, \zeta_N)$, (RCA-OPT:FF) can be directly solved as established in the following theoretical result.

Theorem 3. *The solution of the optimization problem formulation (RCA-OPT:FF) is given by the optimal resource capacity vector $\hat{\boldsymbol{\mu}}^* := (\hat{\mu}_1^*, \dots, \hat{\mu}_N^*)$ where*

$$\hat{\mu}_i^* = \gamma_i + (\mathcal{C} - \langle \mathbf{c}, \boldsymbol{\gamma} \rangle) \frac{\sqrt{\omega_i \zeta_i / c_i}}{\sum_{j=1}^N \sqrt{\omega_j \zeta_j c_j}}. \quad (40)$$

Proof. The Lagrangian associated with (RCA-OPT:FF) is defined as

$$\sum_{i=1}^N \frac{\omega_i \zeta_i}{\hat{\mu}_i - \gamma_i} + \chi \left(\sum_{i=1}^N c_i \hat{\mu}_i - \mathcal{C} \right), \quad (41)$$

where χ is the Lagrange multiplier. Taking the derivative of (41) with respect to $\hat{\mu}_i$ yields

$$\chi c_i - \frac{\omega_i \zeta_i}{(\hat{\mu}_i - \gamma_i)^2},$$

from which upon setting the result to zero and solving for $\hat{\mu}_i$ we obtain

$$\hat{\mu}_i = \gamma_i + \frac{1}{\sqrt{\chi}} \sqrt{\frac{\omega_i \zeta_i}{c_i}}. \quad (42)$$

Substituting (42) into (39) set to equality then renders

$$\sum_{i=1}^N c_i \hat{\mu}_i = \sum_{i=1}^N \gamma_i c_i + \frac{1}{\sqrt{\chi}} \sum_{i=1}^N \sqrt{\omega_i \zeta_i c_i} = \mathcal{C},$$

from which we have

$$\frac{1}{\sqrt{\chi}} = \frac{\mathcal{C} - \sum_{i=1}^N \gamma_i c_i}{\sum_{i=1}^N \sqrt{\omega_i \zeta_i c_i}}. \quad (43)$$

Upon substituting (43) into (42), we obtain the desired result

$$\hat{\mu}_i^* = \gamma_i + \left(\mathcal{C} - \sum_{j=1}^N \gamma_j c_j \right) \frac{\sqrt{\omega_i \zeta_i / c_i}}{\sum_{j=1}^N \sqrt{\omega_j \zeta_j c_j}}.$$

□

7 Introduction

We consider quantum-centric supercomputing environments that comprise a combination of both quantum computing (QC) and high-performance computing (HPC).

{Asser: Background material: Expand}

- Integrating quantum computing and HPC. [8, 24, 39, 40]
- Estimating quantum execution time. [23]
- Several quantum algorithms are promising to solve problems more efficiently than possible on classical computers. However, various complex pre- and post-processing tasks have to be performed when executing a quantum circuit, which requires immense mathematical and technical knowledge. [43]
- Patterns of execution. [15]
- Hybrid classical-quantum optimization. [2]

8 General Case III

Now we consider the application workload model of Section ?? while relaxing the system model assumption that each job reserves the system CPU, QPU and GPU throughout its execution without any overlap with other jobs that wait in the system queue.

For arriving jobs that belong to the sequential-parallel application class comprising a sequential computational phase followed by a parallel computational phase, which occurs with probability $1 - p_{\text{u}}$, we have a stochastic tandem queueing network where the sequential computational phase has a product-form solution — refer to Section ?. This is the case irrespective of whether the sequential computational phase comprises a quantum computational task executed on the system QPU followed by a classical computational task executed on the system GPU, which occurs with probability p_{o} , or a classical computational task executed on the system GPU followed by a quantum computational task executed on the system QPU, which occurs with probability $1 - p_{\text{o}}$. Moreover, from the BCMP theorem [6], we have that the stochastic departure process from the sequential computational phase is Poisson and thus the parallel computational phase can be solved as the corresponding fork-join queueing

network — refer to Section ?? . The solutions of each of these phases can then be combined to obtain the performance of jobs of this type arriving with probability $1 - p_{||}$.

For arriving jobs that belong to the parallel-sequential application class comprising a parallel computational phase followed by a sequential computational phase, which occurs with probability $p_{||}$, we have a fork-join queueing network that can be solved as discussed in Section ?? . However, the departure process from this fork-join queueing network is no longer Poisson and therefore the sequential computational phase no longer has a product-form solution. We likely need to develop an approximation — one option could be to exploit the general approach in [12] which treats cases exactly like this.

For each of these simplified settings, we assume that the computational requirements of nodes “a” and “e” on the left-hand side of Figure ?? are null.

References

- [1] 2007.
- [2] A. Ajagekar, K. A. Hamoud, and F. You. Hybrid classical-quantum optimization techniques for solving mixed-integer programming problems in production scheduling. *IEEE Transactions on Quantum Engineering*, 3:1–16, 2022.
- [3] S. Asmussen. *Applied Probability and Queues*. Springer, second edition, 2003.
- [4] S. Asmussen and P. W. Glynn. *Stochastic Simulation: Algorithms and Analysis*. Springer, 2007.
- [5] S. Balsamo, L. Donatiello, and N. Van Dijk. Bound performance models of heterogeneous parallel processing systems. *IEEE Transactions on Parallel and Distributed Systems*, 9(10):1041–1056, 1998.
- [6] F. Baskett, K. M. Chandy, R. R. Muntz, and F. G. Palacios. Open, closed, and mixed networks of queues with different classes of customers. *Journal of the Association for Computing Machinery*, 22(2):248–260, 1975.
- [7] N. G. Bean, J.-M. Li, and P. G. Taylor. Causal characteristics of qbds with decomposable phase spaces. In G. Latouche and P. Taylor, editors, *Advances in Algorithmic Methods for Stochastic Models*, pages 37–55. Notable Publications, 2000.
- [8] T. Beck, A. Baroni, R. Bennink, G. Buchs, E. A. C. Pérez, M. Eisenbach, R. F. da Silva, M. G. Meena, K. Gottiparthi, P. Groszkowski, T. S. Humble, R. Landfield, K. Maheshwari, S. Oral, M. A. Sandoval, A. Shehata, I.-S. Suh, and C. Zimmer. Integrating quantum computing resources into scientific hpc ecosystems. *Future Generation Computer Systems*, 161:11–25, 2024.

- [9] P. Brémaud. *Markov Chains: Gibbs Fields, Monte Carlo Simulation and Queues*. Springer, second edition, 2020.
- [10] P. J. Burke. The output of a queueing system. *Operations Research*, 4(6):699–704, 1956.
- [11] A. B. Dieker, S. Ghosh, and M. S. Squillante. Optimal resource capacity management for stochastic networks. *Operations Research*, 65(1):221–241, 2017.
- [12] A. B. Dieker and M. S. S. Soumyadip Ghosh. Optimal resource capacity management for stochastic networks. *Operations Research*, 65(1):221–241, 2017.
- [13] L. Flatto. Two parallel queues created by arrivals with two demands II. *SIAM Journal of Applied Mathematics*, 45:861–878, 1985.
- [14] L. Flatto and S. Hahn. Two parallel queues created by arrivals with two demands I. *SIAM Journal of Applied Mathematics*, 44:1041–1053, 1984.
- [15] D. Georg, J. Barzen, M. Beisel, F. Leymann, J. Obst, D. Vietz, B. Weder, and V. Yusupov. Execution patterns for quantum applications. In *ICSOF*T, pages 258–268, 2023.
- [16] J. M. Harrison. *Brownian Motion and Stochastic Flow Systems*. J. Wiley and Sons, 1985.
- [17] J. R. Jackson. Jobshop-like queueing systems. *Management Science*, 10(1):131–142, 1963.
- [18] F. P. Kelly. *Reversibility and Stochastic Networks*. Wiley, 1979.
- [19] H. J. Kushner and G. G. Yin. *Stochastic Approximation and Recursive Algorithms and Applications*. Springer, second edition, 2003.
- [20] G. Latouche and V. Ramaswami. *Introduction to Matrix Analytic Methods in Stochastic Modeling*. ASA-SIAM, Philadelphia, 1999.
- [21] S. S. Lavenberg, editor. *Computer Performance Modeling Handbook*. Academic Press, 1983.
- [22] D. A. Levin, Y. Peres, and E. L. Wilmer. *Markov Chains and Mixing Times*. American Mathematical Society, second edition, 2017.
- [23] N. Ma and H. Li. Understanding and estimating the execution time of quantum circuits, 2025.
- [24] P. Mantha, F. J. Kiwit, N. Saurabh, S. Jha, and A. Luckow. Pilot-quantum: A quantum-hpc middleware for resource, workload and task management, 2025.
- [25] M. Mohanty, G. Gautam, V. Aggarwal, and P. Parag. Analysis of fork-join scheduling on heterogeneous parallel servers. *IEEE/ACM Transactions on Networking*, 32(6):4798–4809, 2024.

- [26] R. R. Muntz. Poisson departure process and queueing networks. Technical Report RC 4145, IBM Research, 1972.
- [27] R. D. Nelson and A. N. Tantawi. Approximate analysis of fork/join synchronization in parallel queues. *IEEE Transactions on Computers*, 37(6):739–743, June 1988.
- [28] M. F. Neuts. Probability distributions of phase type. In *Liber Amicorum Prof. Emeritus H. Florin*, Department of Mathematics, University of Louvain, Belgium, pages 173–206, 1975.
- [29] M. F. Neuts. *Matrix-Geometric Solutions in Stochastic Models: An Algorithmic Approach*. 1981.
- [30] M. F. Neuts. *Matrix-Geometric Solutions in Stochastic Models: An Algorithmic Approach*. The Johns Hopkins University Press, 1981.
- [31] M. F. Neuts. The caudal characteristic curve of queues. *Advances in Applied Probability*, 18:221–254, 1986.
- [32] M. F. Neuts. *Structured Stochastic Matrices of M/G/1 Type and Their Applications*. Marcel Dekker, 1989.
- [33] V. Nguyen. The trouble with diversity: Fork-join networks with heterogeneous customer population. *Ann. Appl. Probab.*, 4(4):1–25, 1994.
- [34] M. I. Reiman and B. Simon. An interpolation approximation for queueing systems with poisson input. *Operations Research*, 36(3):454–469, 1988.
- [35] M. I. Reiman and B. Simon. Light traffic limits of sojourn time distributions in markovian queueing networks. *Stochastic Models*, 4(2):191–233, 1988.
- [36] M. I. Reiman and B. Simon. Open queueing systems in light traffic. *Mathematics of Operations Research*, 14(1):26–59, February 1989.
- [37] D. Schol, M. Vlasiov, and B. Zwart. Extreme values for the waiting time in large fork-join queues. *Queueing Systems*, 2025.
- [38] E. Seneta. *Non-Negative Matrices and Markov Chains*. Springer Verlag, second edition, 1981.
- [39] A. Shehata et al. Bridging paradigms: Designing for HPC-Quantum convergence. *Future Generation Computer Systems*, 174:107980, 2026.
- [40] M. Slys, P. Rydlichowski, K. Kurowski, O. Bacarreza, E. C. Gomez, Z. Chandani, B. Heim, P. Khalate, W. R. Clements, and J. Fletcher. Hybrid classical-quantum supercomputing: A demonstration of a multi-user, multi-qpu and multi-gpu environment, 2025.

- [41] S. Varma. *Heavy and light traffic approximations for queues with synchronization constraints*. PhD thesis, University of Maryland, College Park, MD, USA, 1990.
- [42] S. Varma and A. M. Makowski. Interpolation approximations for symmetric fork-join queues. *Performance Evaluation*, 20:245–265, 1994.
- [43] B. Weder, U. Breitenbücher, F. Leymann, and K. Wild. Integrating quantum computing into workflow modeling and execution. In *2020 IEEE/ACM 13th International Conference on Utility and Cloud Computing (UCC)*, pages 279–291, 2020.
- [44] W. Whitt. *Stochastic-Process Limits: An Introduction to Stochastic-Process Limits and Their Application to Queues*. Springer, 2002.
- [45] R. W. Wolff. *Stochastic Modeling and the Theory of Queues*. Prentice Hall, 1989.

A Light-Heavy Traffic Interpolation Approximation

The light-heavy traffic interpolation approximation can be an effective method for analyzing otherwise intractable queueing systems. The central idea is to approximate some quantity $f(\lambda)$ of interest in the system based on a combination of n light-traffic derivatives associated with the Taylor series expansion of $f(\lambda)$ at $\lambda = 0$ and the heavy-traffic limit of $f(\lambda)$ as λ approaches its critical value μ^* . More specifically, following the approach developed by Reiman and Simon [34–36], we first scale $f(\lambda)$ by defining

$$g(\lambda) := (\mu^* - \lambda)f(\lambda), \quad \lambda \in [0, \mu^*), \quad (44)$$

and then we interpolate $g(\lambda)$ by a polynomial of order $n + 1$ of the form

$$\tilde{g}(\lambda) = a_{n+1}\lambda^{n+1} + a_n\lambda^n + \dots + a_1\lambda + a_0, \quad (45)$$

where $a_{n+1}, a_n, \dots, a_1, a_0$ are determined from the n light-traffic derivatives associated with the Taylor series expansion of $f(\lambda)$ at $\lambda = 0$, i.e.,

$$f(\lambda) \approx f^{(0)}(0) + \lambda f^{(1)}(0) + \dots + \frac{\lambda^{n-1}}{(n-1)!} f^{(n-1)}(0) + \frac{\lambda^n}{n!} f^{(n)}(0), \quad \lambda \approx 0, \quad (46)$$

together with the heavy-traffic limit

$$\lim_{\lambda \rightarrow \mu^*} (\mu^* - \lambda)f(\lambda) = h. \quad (47)$$

Here, assuming the existence of the first n derivatives of $f(\lambda)$ at $\lambda = 0$, the $f^{(k)}(0)$ in (46) denotes the order k derivative of $f(\lambda)$ at $\lambda = 0$, namely

$$f^{(k)}(0) = \left. \frac{d^k f}{d\lambda^k} \right|_{\lambda=0}, \quad k \in \{0, 1, \dots, n\}.$$

From (46) and (47), we have the matching light-traffic conditions

$$\tilde{g}^{(k)}(0) = g^{(k)}(0), \quad k \in \{0, 1, \dots, n\} \quad (48)$$

and the heavy-traffic condition

$$\tilde{g}(\mu^*) = g(\mu^*) = h. \quad (49)$$

Then, once $\tilde{g}(\lambda)$ is determined, we invert the scaling in (44) to obtain the interpolation approximation of $f(\lambda)$ as

$$\tilde{f}(\lambda) = \frac{\tilde{g}(\lambda)}{\mu^* - \lambda}, \quad \lambda \in [0, \mu^*). \quad (50)$$

The calculation of the derivatives at zero for the quantity of interest $f(\lambda)$ is clearly a focal point of the light-heavy traffic interpolation approximation approach. This is made possible within the context of Poisson (or Poisson-driven) arrival processes by the light-traffic theory rigorously developed in [36], where the authors furnish expressions that, in principle, enable us to calculate the derivative of any order n . The availability of the light traffic derivatives $f^{(k)}(0)$ for all k fully determines $f(\lambda)$ in terms of its Taylor series expansion, and thus provides a way to obtain an exact expression $\tilde{f}(\lambda) = f(\lambda)$. However, in our particular situation as well as in most practical cases, the calculations quickly become extremely involved and we therefore have to restrict ourselves to the 0^{th} and 1^{st} order derivatives. Nevertheless, we are strongly encouraged by the fact that our light-heavy traffic interpolation approximation for the complex unbalanced application workload with nonhomogeneous service rates $\hat{\mu}_1$ and $\hat{\mu}_2$ recovers the exact expression (3) for the expected sojourn time under the balanced workload with homogeneous service rates $\hat{\mu} = \hat{\mu}_1 = \hat{\mu}_2$ due to Nelson and Tantawi [27]. Moreover, an a posteriori comparison of the resulting approximations with simulations strongly suggests that our quadratic polynomial interpolation in λ is an effective approximation.

Our interests are in deriving a light-heavy traffic interpolation approximation for the more difficult problem of approximating the expected sojourn time of jobs in a fork-join queueing network comprising two parallel servers under unbalanced application workloads in which the arrivals follow a Poisson process with rate γ and the nonhomogeneous service times for each of the two servers respectively follow an exponential distribution with rates $\hat{\mu}_1$ and $\hat{\mu}_2$. The quantity of interest in the system is therefore $f(\lambda) = \mathbb{E}[T_{FJ}]$ whose approximation $\tilde{f}(\lambda) = \mathbb{E}[\tilde{T}_{FJ}]$ is based on a combination of the 0^{th} and 1^{st} order derivatives $f^{(0)}(0)$ and $f^{(1)}(0)$ associated with the Taylor series expansion of $f(\lambda)$ at $\lambda = 0$ in (46) and the heavy-traffic limit of $f(\lambda)$ as λ approaches its critical value $\mu^* := \min\{\hat{\mu}_1, \hat{\mu}_2\}$ in (47). We note that the fork-join queueing network of interest satisfies the relevant technical conditions required by the light-heavy traffic interpolation approximation approach of Reiman and Simon [34–36, 41].

Following [34–36], consider the sojourn time T_{FJ} of a tagged job that arrives at time $t = 0$ whose service times v_1 and v_2 are random variables that follow an exponential distribution with mean $\mathbb{E}[v_1] = \hat{\mu}_1$ and $\mathbb{E}[v_2] = \hat{\mu}_2$, respectively, where v_1 and v_2 are independent of each other as well as all other random variables of the fork-join queueing network. The 0^{th} order derivative can be obtained by definition as

$$f^{(0)}(0) = \mathbb{E}[\psi(\emptyset)] := \mathbb{E}[T_{FJ} \mid \text{no arrivals over } (-\infty, 0)], \quad (51)$$

noting that any arrival at time $t \in (0, \infty)$ will be serviced after the tagged job and therefore not impact T_{FJ} . We then have the following theoretical result for $f^{(0)}(0)$.

Lemma 1. *The 0th order derivative of the fork-join queueing network is given by*

$$f^{(0)}(0) = \mathbb{E}[\psi(\emptyset)] = \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}. \quad (52)$$

Proof. Since the tagged job does not encounter any prior arrivals, we have by definition that $\mathbb{E}[\psi(\emptyset)] = \mathbb{E}[\max\{v_1, v_2\}]$ where the random variable $\psi(\emptyset) = \max\{v_1, v_2\}$ follows a phase-type distribution $(\boldsymbol{\alpha}, \mathbf{Q}_{FJ})$ [20, 28, 30] defined on a four-state Markov process with infinitesimal generator matrix $\mathbf{Q}_{FJ}$ is expressed as

$$\mathbf{Q}_{FJ} = \begin{bmatrix} \tilde{\mathbf{Q}} & \tilde{\mathbf{q}} \\ \mathbf{0} & 0 \end{bmatrix}, \quad \tilde{\mathbf{Q}} = \begin{bmatrix} -(\hat{\mu}_1 + \hat{\mu}_2) & \hat{\mu}_1 & \hat{\mu}_2 \\ 0 & -\hat{\mu}_2 & 0 \\ 0 & 0 & -\hat{\mu}_1 \end{bmatrix}, \quad \tilde{\mathbf{q}} = \begin{bmatrix} 0 \\ \hat{\mu}_2 \\ \hat{\mu}_1 \end{bmatrix}, \quad (53)$$

together with the initial probability vector $(\boldsymbol{\alpha}, \alpha_{00})$ where $\boldsymbol{\alpha} = [1, 0, 0]$ and $\alpha_{00} = 0$. Defining $\exp(\mathbf{Y}) := \sum_{j=0}^{\infty} \frac{1}{j!} \mathbf{Y}^j$ for any given matrix $\mathbf{Y}$, the distribution function $H(\cdot)$ of the random variable $\psi(\emptyset)$ is given by

$$H(y) = 1 - \boldsymbol{\alpha} \exp(\tilde{\mathbf{Q}}y) \mathbf{e}, \quad y \in \mathbb{R}_{\geq 0},$$

from which we have $\mathbb{E}[\psi(\emptyset)] = -\boldsymbol{\alpha} \tilde{\mathbf{Q}}^{-1} \mathbf{e}$. Then we obtain from (53) and direct calculation

$$\tilde{\mathbf{Q}}^{-1} = \begin{bmatrix} -\frac{1}{\hat{\mu}_1 + \hat{\mu}_2} & -\frac{\hat{\mu}_1}{\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)} & -\frac{\hat{\mu}_2}{\hat{\mu}_1(\hat{\mu}_1 + \hat{\mu}_2)} \\ 0 & -\frac{1}{\hat{\mu}_2} & 0 \\ 0 & 0 & -\frac{1}{\hat{\mu}_1} \end{bmatrix},$$

which together with $-\boldsymbol{\alpha} \tilde{\mathbf{Q}}^{-1} \mathbf{e}$ yields the desired result (52). $\square$

Turning to the 1st order derivative, we consider the sojourn time T_{FJ} of a tagged job that arrives at time 0 with exponential service times v_1 and v_2 as above, given that another job arrives at time $t \in (-\infty, \infty)$ whose service times w_1 and w_2 are random variables that follow an exponential distribution with mean $\mathbb{E}[w_1] = \hat{\mu}_1$ and $\mathbb{E}[w_2] = \hat{\mu}_2$, respectively, where v_1, v_2, w_1, w_2 are independent of each other as well as all other random variables of the fork-join queueing network. The 1st order derivative can be obtained by definition as

$$\mathbb{E}[\psi(\{t\})] := \mathbb{E}[T_{FJ} \mid \text{exactly one arrival at time } t \text{ over } (-\infty, \infty)], \quad (54)$$

$$f^{(1)}(0) = \int_{-\infty}^{\infty} (\mathbb{E}[\psi(\{t\})] - \mathbb{E}[\psi(\emptyset)]) dt. \quad (55)$$

We then have the following theoretical result for $f^{(1)}(0)$.

Lemma 2. *The 1st order derivative of the fork-join queueing network is given by*

$$f^{(1)}(0) = \frac{1}{\hat{\mu}_1^2} + \frac{1}{\hat{\mu}_2^2} - \frac{2}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4}. \quad (56)$$

Proof. We have by definition that

$$\psi(\{t\}) = \begin{cases} \max\{v_1, v_2\}, & t \in [0, \infty), \\ \max\{v_1 + (w_1 + t)^+, v_2 + (w_2 + t)^+\}, & t \in (-\infty, 0), \end{cases}$$

where $(x)^+ := \max\{0, x\}$, from which together with (55) we conclude

$$f^{(1)}(0) = \int_{-\infty}^0 (\mathbb{E}[\psi(\{t\})] - \mathbb{E}[\psi(\emptyset)]) dt, \quad (57)$$

given that $\mathbb{E}[\psi(\{t\})] = \mathbb{E}[\max\{v_1, v_2\}] = \mathbb{E}[\psi(\emptyset)]$ for $t \in [0, \infty)$. By definition, the random variables $v_1 + (w_1 + t)^+$ and $v_2 + (w_2 + t)^+$ for fixed $t \in (-\infty, 0)$ are independent and respectively follow the distribution function given by

$$\begin{aligned} H_1(y) &= 1 - \hat{H}_1(y), & H_2(y) &= 1 - \hat{H}_2(y), & y &\in \mathbb{R}_{\geq 0}, \\ \hat{H}_1(y) &= (1 + \hat{\mu}_1 y \exp(\hat{\mu}_1 t)) \exp(-\hat{\mu}_1 y), & \hat{H}_2(y) &= (1 + \hat{\mu}_2 y \exp(\hat{\mu}_2 t)) \exp(-\hat{\mu}_2 y). \end{aligned}$$

From the independence of these two random variables, we obtain for fixed $t \in (-\infty, 0)$

$$\begin{aligned} \mathbb{E}[\psi(\{t\})] &= \int_0^\infty (1 - H_1(y)H_2(y)) dy \\ &= \int_0^\infty \hat{H}_1(y) dy + \int_0^\infty \hat{H}_2(y) dy - \int_0^\infty \hat{H}_1(y)\hat{H}_2(y) dy, \\ &= \frac{1 + e^{\hat{\mu}_1 t}}{\hat{\mu}_1} + \frac{1 + e^{\hat{\mu}_2 t}}{\hat{\mu}_2} - \frac{1}{\hat{\mu}_1 + \hat{\mu}_2} - \frac{\hat{\mu}_1 e^{\hat{\mu}_1 t} + \hat{\mu}_2 e^{\hat{\mu}_2 t}}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1\hat{\mu}_2 e^{(\hat{\mu}_1 + \hat{\mu}_2)t}}{(\hat{\mu}_1 + \hat{\mu}_2)^3}. \end{aligned} \quad (58)$$

Upon substituting (58) and (52) from Lemma 1 into (57), we then derive

$$\begin{aligned} f^{(1)}(0) &= \int_{-\infty}^0 \left(\frac{e^{\hat{\mu}_1 t}}{\hat{\mu}_1} + \frac{e^{\hat{\mu}_2 t}}{\hat{\mu}_2} - \frac{\hat{\mu}_1 e^{\hat{\mu}_1 t} + \hat{\mu}_2 e^{\hat{\mu}_2 t}}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1\hat{\mu}_2 e^{(\hat{\mu}_1 + \hat{\mu}_2)t}}{(\hat{\mu}_1 + \hat{\mu}_2)^3} + \frac{1}{\hat{\mu}_1} + \frac{1}{\hat{\mu}_2} \right. \\ &\quad \left. - \frac{1}{\hat{\mu}_1 + \hat{\mu}_2} - \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)} \right) dt, \\ &= \int_{-\infty}^0 \left(\frac{e^{\hat{\mu}_1 t}}{\hat{\mu}_1} + \frac{e^{\hat{\mu}_2 t}}{\hat{\mu}_2} - \frac{\hat{\mu}_1 e^{\hat{\mu}_1 t} + \hat{\mu}_2 e^{\hat{\mu}_2 t}}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1\hat{\mu}_2 e^{(\hat{\mu}_1 + \hat{\mu}_2)t}}{(\hat{\mu}_1 + \hat{\mu}_2)^3} \right) dt, \\ &= \frac{1}{\hat{\mu}_1^2} + \frac{1}{\hat{\mu}_2^2} - \frac{2}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4}, \end{aligned}$$

thus yielding the desired result. $\square$

Next, we consider the heavy-traffic limit of $f(\lambda)$ in (47) as λ approaches its critical value $\mu^* := \min\{\hat{\mu}_1, \hat{\mu}_2\}$. Without loss of generality, suppose that $\hat{\mu}_1 \leq \hat{\mu}_2$. We first observe that, in the limit as $(\hat{\mu}_2 - \hat{\mu}_1) \rightarrow 0$, the application workload of the fork-join queueing network becomes balanced and thus the heavy-traffic limit is known to be $h = 11/8$ [42]. On the other hand, in the limit as $(\hat{\mu}_2 - \hat{\mu}_1) \rightarrow \infty$, or equivalently $\hat{\mu}_2^{-1} \rightarrow 0$, the application workload of the fork-join queueing network becomes unbalanced to the point of reducing the queueing network to the corresponding M/M/1 queue with service rate $\hat{\mu}_1$ in which case the heavy-traffic limit is known to be $h = 1$ [16, 44]. The heavy-traffic limit for the fork-join queueing network under an unbalanced workload is therefore within the interval $h \in [1, 11/8]$. We then have the following theoretical result for the heavy-traffic limit h under any given $\hat{\mu}_1$ and $\hat{\mu}_2$

Theorem 4. *The heavy-traffic limit of the fork-join queueing network is given by*

$$\lim_{\lambda \rightarrow \mu^*} (\mu^* - \lambda)f(\lambda) = h = TBA. \quad (59)$$

Proof. WORK OUT THE DETAILS □

We now exploit the above results to obtain the desired light-heavy traffic interpolation approximation $\tilde{f}(\lambda)$ for the expected sojourn time of jobs in the two-server fork-join queueing network of interest under unbalanced application workloads.

Theorem 5. *Consider the sojourn time T_{FJ} of jobs in a fork-join queueing network comprising two parallel servers where arrivals follow a Poisson process with rate γ and service times follow an exponential distribution with nonhomogeneous rates $\hat{\mu}_1$ and $\hat{\mu}_2$ at server 1 and 2, respectively. Further let $f(\lambda) = \mathbb{E}[T_{FJ}]$ and, without loss of generality, $\hat{\mu}_1 = \mu^* := \min\{\hat{\mu}_1, \hat{\mu}_2\}$. The light-heavy traffic interpolation approximation $\tilde{f}(\lambda)$ for $f(\lambda)$ is then given by*

$$\tilde{f}(\lambda) = \frac{\lambda}{\hat{\mu}_1 - \lambda} \left(\frac{1}{\hat{\mu}_1} + \frac{\hat{\mu}_1}{\hat{\mu}_2^2} - \frac{2\hat{\mu}_1}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1^2\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4} \right) + \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}, \quad \lambda \in [0, \hat{\mu}_1), \quad (60)$$

Proof. The combination of (44), (46) and (47) yields

$$g(\lambda) = (\hat{\mu}_1 - \lambda) \left(f^{(0)}(0) + \lambda f^{(1)}(0) + \frac{\lambda^2}{2} h \right),$$

from which together with (52) in Lemma 1, (56) in Lemma 2 and (59) in Theorem 4 we

derive

$$\begin{aligned}
g(0) &= (\hat{\mu}_1 - 0) \left(\frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)} + 0 \cdot f^{(1)}(0) + 0 \cdot h \right) = \hat{\mu}_1 \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}, \\
g'(0) &= \frac{d}{d\lambda} \left[(\hat{\mu}_1 - \lambda) \left(f^{(0)}(0) + \lambda f^{(1)}(0) + \frac{\lambda^2}{2} h \right) \right] \Big|_{\lambda=0}, \\
&= \frac{d}{d\lambda} \left[\hat{\mu}_1 f^{(0)}(0) + \lambda \hat{\mu}_1 f^{(1)}(0) + \frac{\lambda^2 \hat{\mu}_1}{2} h - \lambda f^{(0)}(0) - \lambda^2 f^{(1)}(0) - \frac{\lambda^3}{2} h \right] \Big|_{\lambda=0}, \\
&= \left[0 + \hat{\mu}_1 f^{(1)}(0) + \lambda \hat{\mu}_1 h - f^{(0)}(0) - 2\lambda f^{(1)}(0) - \frac{3\lambda^2}{2} h \right] \Big|_{\lambda=0}, \\
&= \hat{\mu}_1 f^{(1)}(0) - f^{(0)}(0) = \frac{1}{\hat{\mu}_1} + \frac{\hat{\mu}_1}{\hat{\mu}_2^2} - \frac{2\hat{\mu}_1}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1^2\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4} - \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}, \\
g(\mu^*) &= g(\hat{\mu}_1) = h.
\end{aligned}$$

With respect to (45), the quadratic interpolation $\tilde{g}(\lambda)$ corresponding to $g(\lambda)$ is given by

$$\tilde{g}(\lambda) = a_2\lambda^2 + a_1\lambda + a_0.$$

From the matching light-traffic conditions (48) and the heavy-traffic condition (49) we have

$$\begin{aligned}
a_2 &= 0, \\
a_1 &= \frac{1}{\hat{\mu}_1} + \frac{\hat{\mu}_1}{\hat{\mu}_2^2} - \frac{2\hat{\mu}_1}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1^2\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4} - \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}, \\
a_0 &= \hat{\mu}_1 \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)},
\end{aligned}$$

from which we obtain

$$\begin{aligned}
\tilde{g}(\lambda) &= \lambda \left(\frac{1}{\hat{\mu}_1} + \frac{\hat{\mu}_1}{\hat{\mu}_2^2} - \frac{2\hat{\mu}_1}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1^2\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4} - \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)} \right) + \hat{\mu}_1 \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}, \\
&= \lambda \left(\frac{1}{\hat{\mu}_1} + \frac{\hat{\mu}_1}{\hat{\mu}_2^2} - \frac{2\hat{\mu}_1}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1^2\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4} \right) + (\hat{\mu}_1 - \lambda) \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}
\end{aligned}$$

Then, given this quadratic interpolation $\tilde{g}(\lambda)$, we obtain from (50) the desired interpolation approximation $\tilde{f}(\lambda)$ of $f(\lambda)$ as

$$\tilde{f}(\lambda) = \frac{\lambda}{\hat{\mu}_1 - \lambda} \left(\frac{1}{\hat{\mu}_1} + \frac{\hat{\mu}_1}{\hat{\mu}_2^2} - \frac{2\hat{\mu}_1}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1^2\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4} \right) + \frac{\hat{\mu}_1 - \lambda}{\hat{\mu}_1 - \lambda} \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}, \quad (61)$$

for $\lambda \in [0, \hat{\mu}_1]$, thus completing the proof. $\square$

Remark 1. It is important to note that, upon substituting $\hat{\mu}_1 = \hat{\mu}_2 = \hat{\mu}$ in (60), our light-heavy traffic interpolation approximation recovers the exact expected sojourn time of jobs in the two-server fork-join queueing network of interest under balanced application workloads, specifically expressed as

$$\mathbb{E}[T_{FJ}] = \frac{12 - \lambda/\hat{\mu}}{8(\hat{\mu} - \lambda)}, \quad (62)$$

due to Nelson and Tantawi [27].

B Optimal Resource Allocation

The objective of our capacity planning optimization problem then is to minimize the expected stationary sojourn times weighted by a vector $\boldsymbol{\omega} := (\omega_1, \dots, \omega_M)$, subject to a budget constraint $\mathcal{C}$ and to the constraint that the stochastic network is stable. More precisely, our capacity planning optimization problem is formulated as follows.

$$\text{(OPT)} \quad \min_{\tilde{\boldsymbol{\mu}} \in (0, \infty)^M} \sum_{i=1}^M \omega_i \mathbb{E}[T_i] \quad (63)$$

$$\text{s.t.} \quad \langle \mathbf{c}, \tilde{\boldsymbol{\mu}} \rangle \leq \mathcal{C}, \quad \text{and} \quad \hat{\mu}_i > \gamma_i, \quad \forall i \in [M]. \quad (64)$$

To determine the optimal capacity allocation vector $\tilde{\boldsymbol{\mu}}$ subject to the constraints (64), the optimization problem (OPT) can be solved via Lagrangian methods. Specifically, the Lagrangian associated with (OPT) is defined as

$$\sum_{i=1}^M \omega_i \mathbb{E}[T_i] + \chi \left(\sum_{i=1}^M c_i \hat{\mu}_i - \mathcal{C} \right), \quad (65)$$

where $\mathbb{E}[T_i]$, $i \in [M-1]$, is given by (2), $\mathbb{E}[T_M]$ is given by (3), and χ is the Lagrange multiplier. Taking the derivative of (65) with respect to $\hat{\mu}_i$ for the first four queues and setting the result to zero, we have

$$\frac{\omega_i}{(\hat{\mu}_i - \gamma_i)^2} = \chi c_i \quad \Rightarrow \quad \hat{\mu}_i = \gamma_i + \frac{1}{\sqrt{\chi}} \sqrt{\frac{\omega_i}{c_i}}. \quad (66)$$

Similarly, taking the derivative of (65) with respect to $\hat{\mu}_M$ for the fork-join queue and setting the result to zero, we obtain

$$\frac{\omega_M(12\hat{\mu}_M^2 - 2\lambda\hat{\mu}_M + \lambda^2)}{8\hat{\mu}_M^2(\hat{\mu}_M - \lambda)^2} = \chi c_M, \quad (67)$$

which after detailed derivations leads to the following fourth-order polynomial, or quartic, equation of $\hat{\mu}_M$:

$$\hat{\mu}_M^4 - 2\lambda\hat{\mu}_M^3 + \left(\lambda^2 - \frac{3\omega_M}{2\chi c_M} \right) \hat{\mu}_M^2 + \frac{\lambda\omega_M}{4\chi c_M} \hat{\mu}_M - \frac{\lambda^2\omega_M}{8\chi c_M} = 0. \quad (68)$$

The solution of our capacity planning optimization problem can then be efficiently computed via known numerical methods together with (65) – (68).